

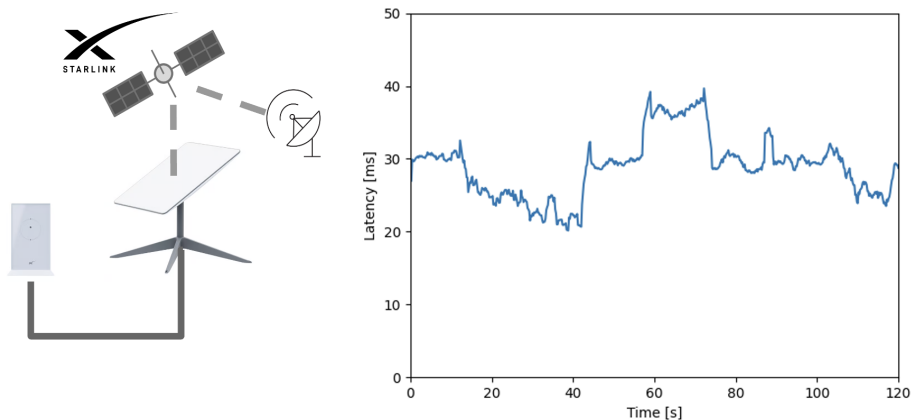
Introduction to Probability and Statistics

Lecture 1 Exercises

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Exercise 1: Use one of the files with data in the moodle materials folder to plot the absolute, relative, and normalized histograms of (choose one):

- a) The latency for communication in our Starlink setup (file: Starlink_latency.csv)



- b) The temperature of a room measured by a sensor during a PhD course at Oulu University (file: temperature.csv)

Try different numbers of bins. Is the data approximately normal?

Exercise 2: Simulate rolling a fair dice n times or roll it yourself.

- Calculate the sample mean $\bar{X}_i$ and variance S_i^2 for all $i = 1, 2, \dots, n$.
- Plot the sample mean $\bar{X}_i$ and $\bar{X}_i \pm S_i$.
- Plot a normalized histogram with the n outcomes. How many times do you need to roll the dice so the histogram resembles the pmf of a uniform RV?

Tip: You can use the code `dice_loln.py` in the moodle page as a base

Exercise 3: Consider the case of flipping a fair coin n times and let H_n be the random variable (RV) of the number of *heads*. Also recall that the sum of n Bernoulli RVs is a Binomial random variable with mean np and variance $np(1-p)$. Calculate and compare the probability of observing $H_{10} = 7$ and $H_{100} = 70$ using each of the following methods.

- a) Using the formula for the probability mass function (pmf) of Binomial RVs

$$P(H_n = k) = \binom{n}{k} p^k (1-p)^{n-k} \quad (1)$$

- b) Using the central limit theorem

Now calculate and compare the probability of observing $H_{10} \geq 7$ and $H_{100} \geq 70$ with the following methods.

- c) Summing over the formula for the pmf of Binomial RVs

$$P(H_n \geq k) = \sum_{i=k}^n \binom{n}{i} p^i (1-p)^{n-i} \quad (2)$$

- d) Using the central limit theorem

Exercise 4: A football team will play 60 games this year. Thirty-two of these games are against teams playing the champions league, denoted as class A teams, and 28 are against other teams, denoted as class B teams. The outcomes of the games are independent. The team will win each game against a class A team with probability 0.5, and it will win each game against a class B team with probability 0.7. Let X denote its total number of victories in the year.

- Is X a Binomial RV?
- Let X_A and X_B denote, respectively, the number of victories against class A and class B teams. What are their distributions?
- What is the relationship between X , X_A and X_B ?
- Approximate the probability that the team wins 40 or more games this year